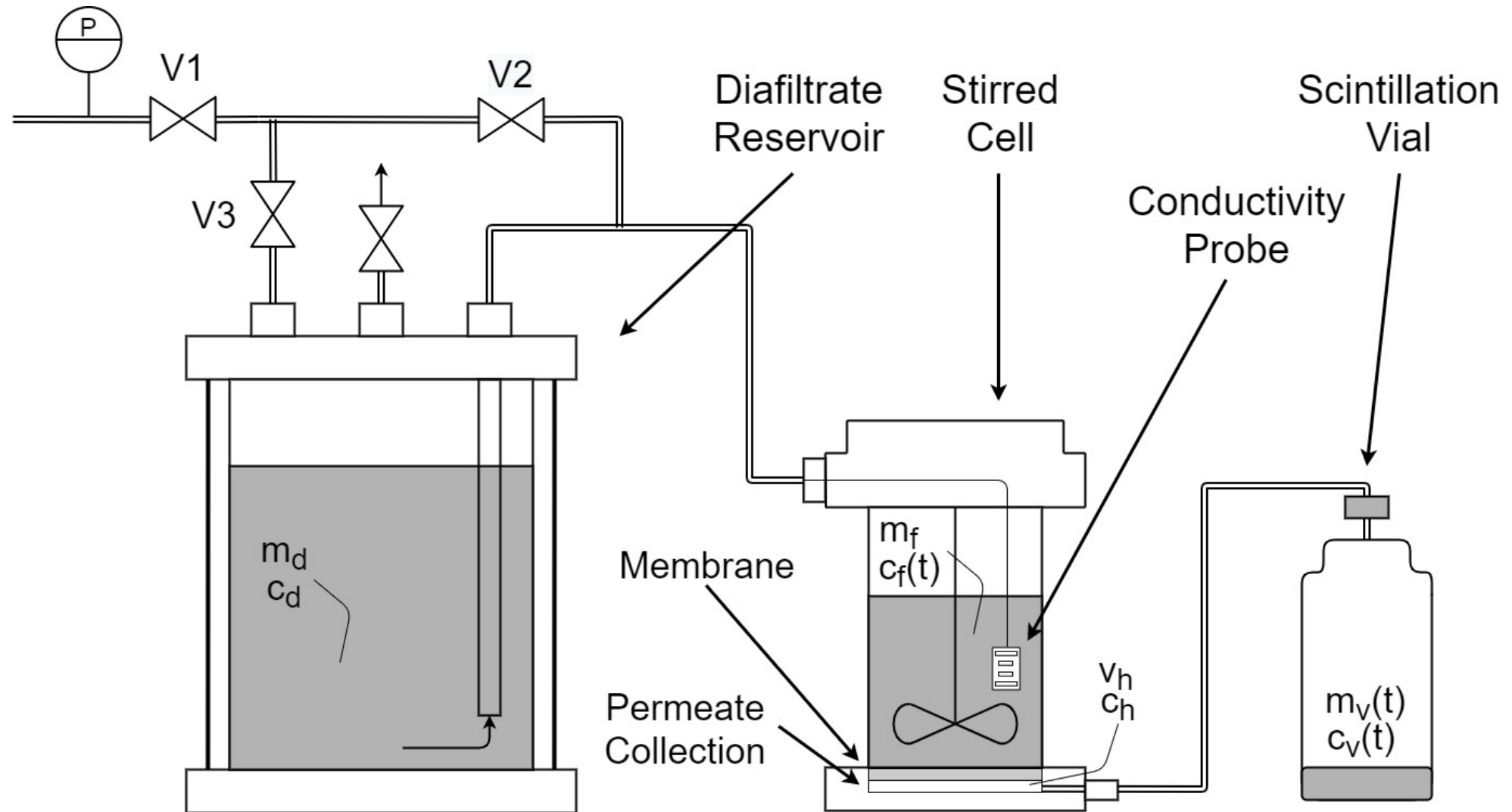


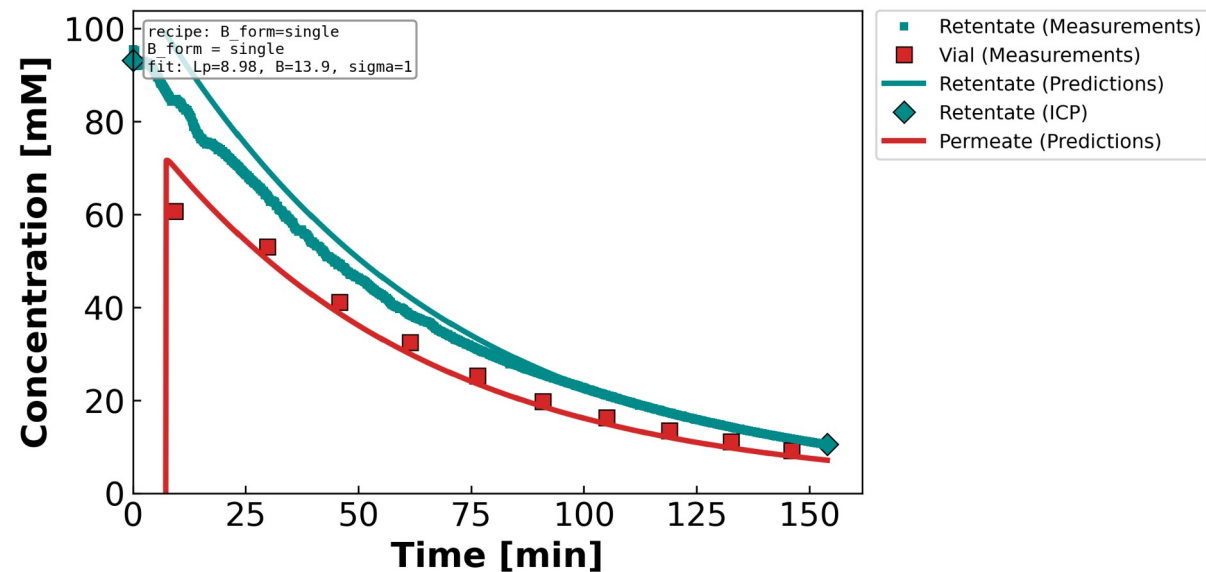
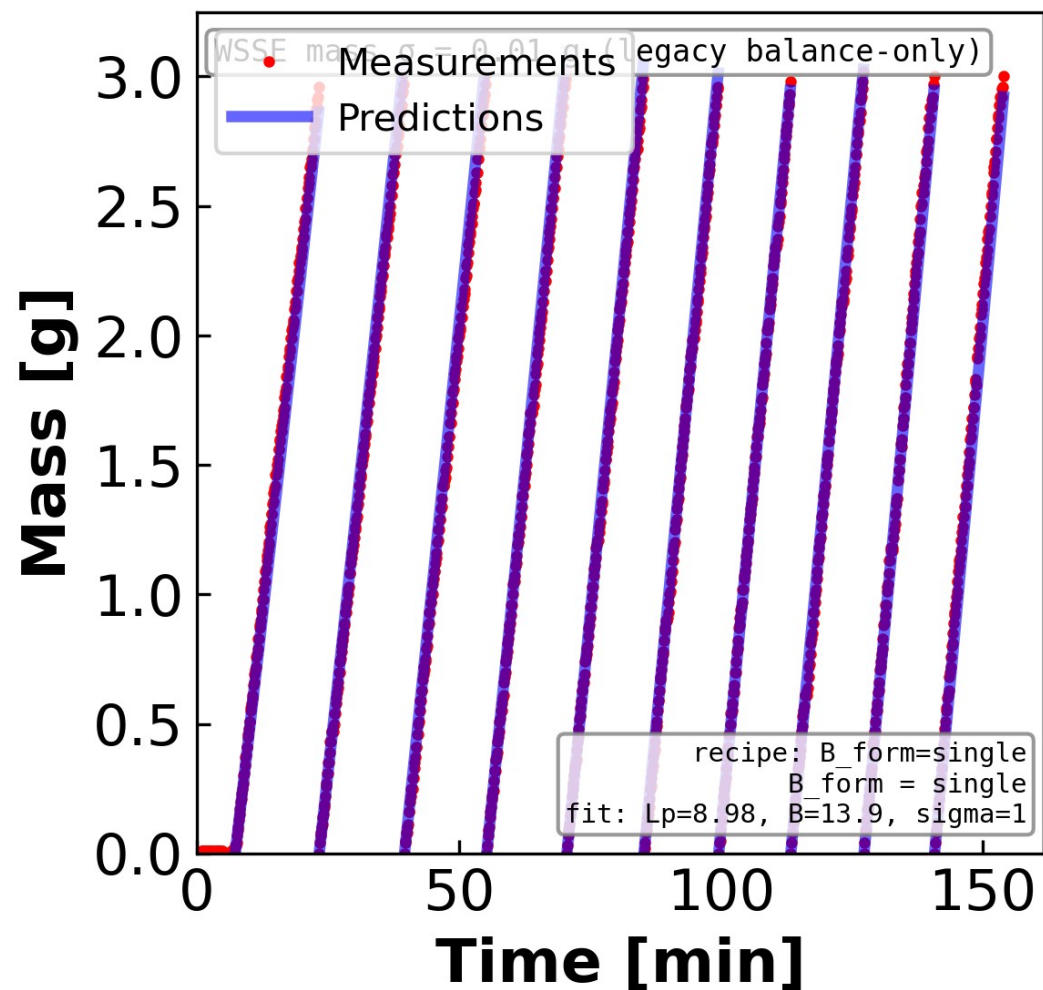
Backup — Supporting Figures

Fits, full $\sigma \times B$ contours, Péclet regression, NaCl B(c), concentration-flux

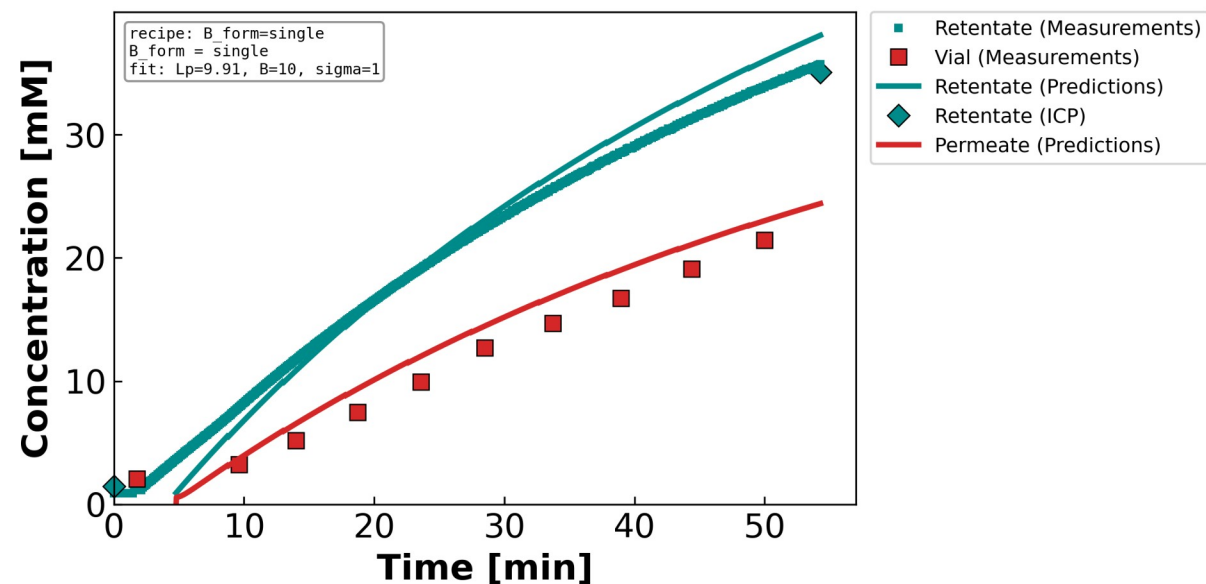
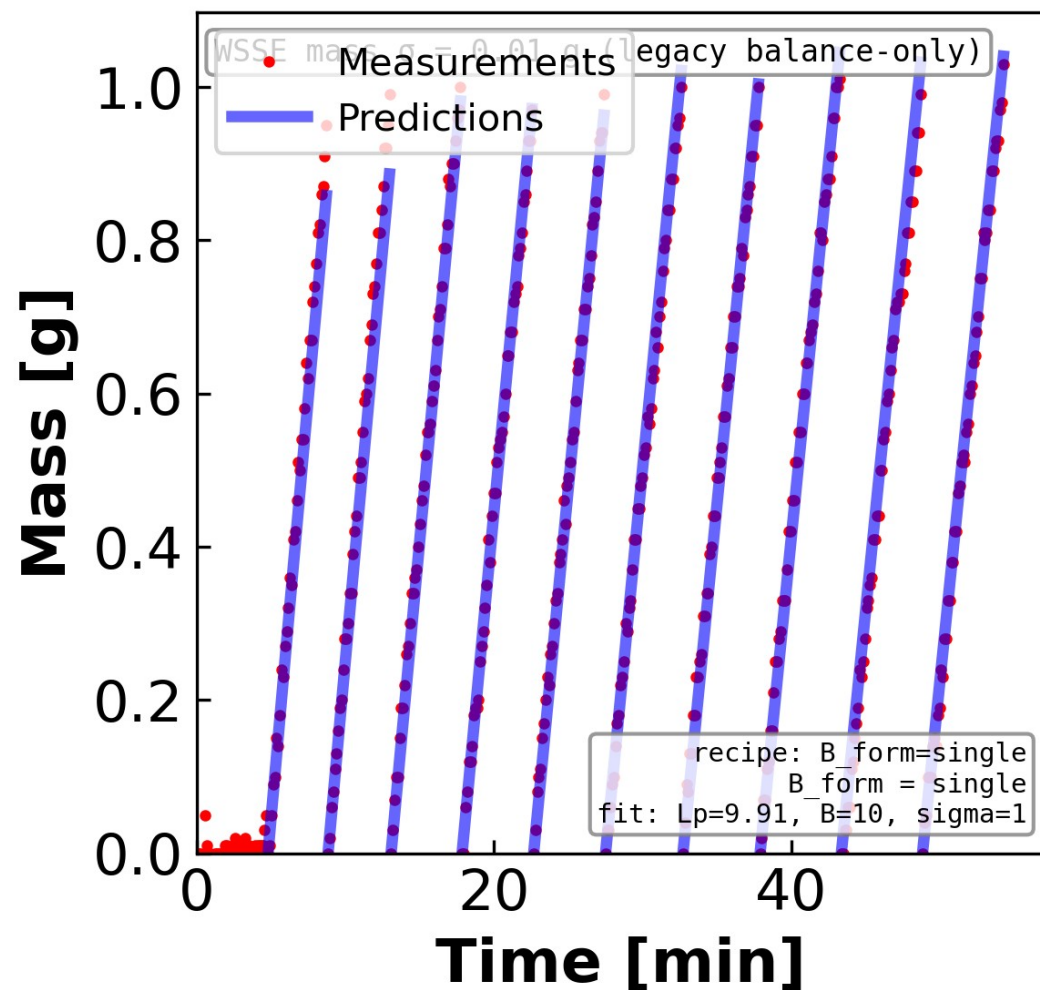
Backup — Physics-Based Model Links Data to Transport Parameters



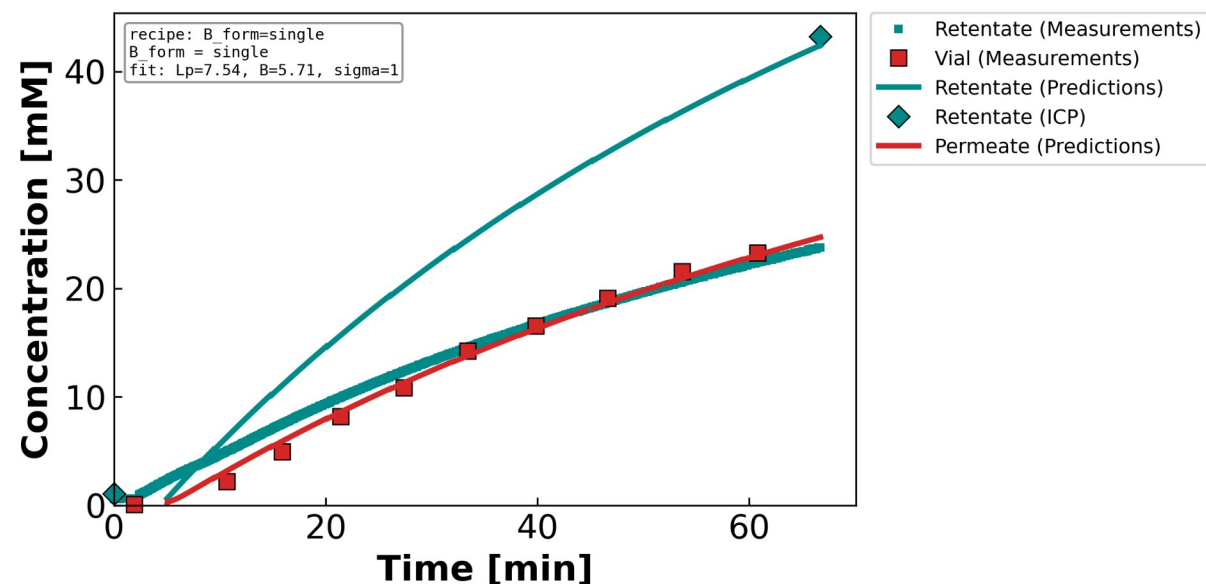
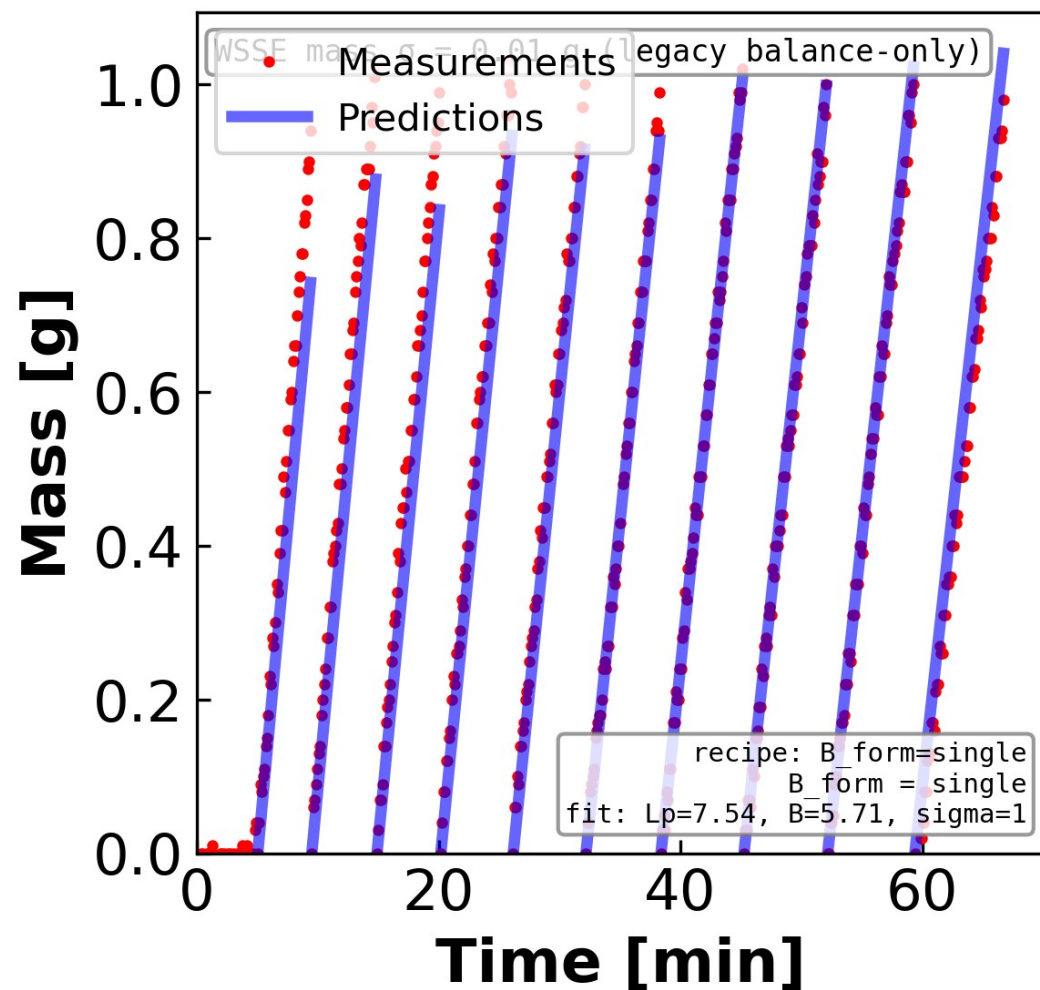
Backup — Figure 1. Measured and model-predicted time profiles for a representative...



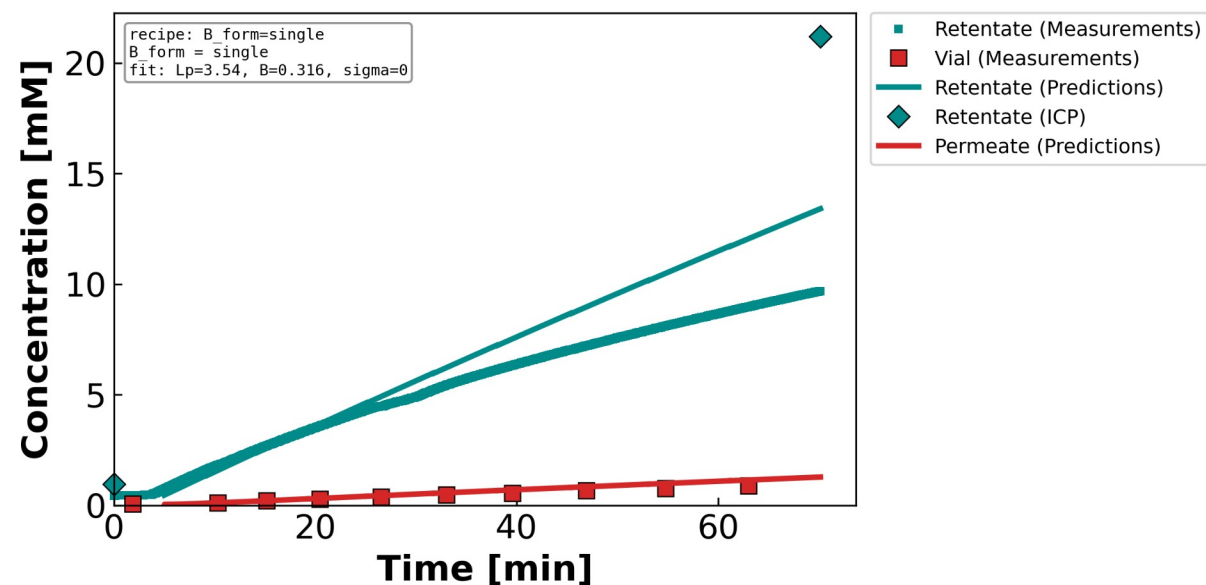
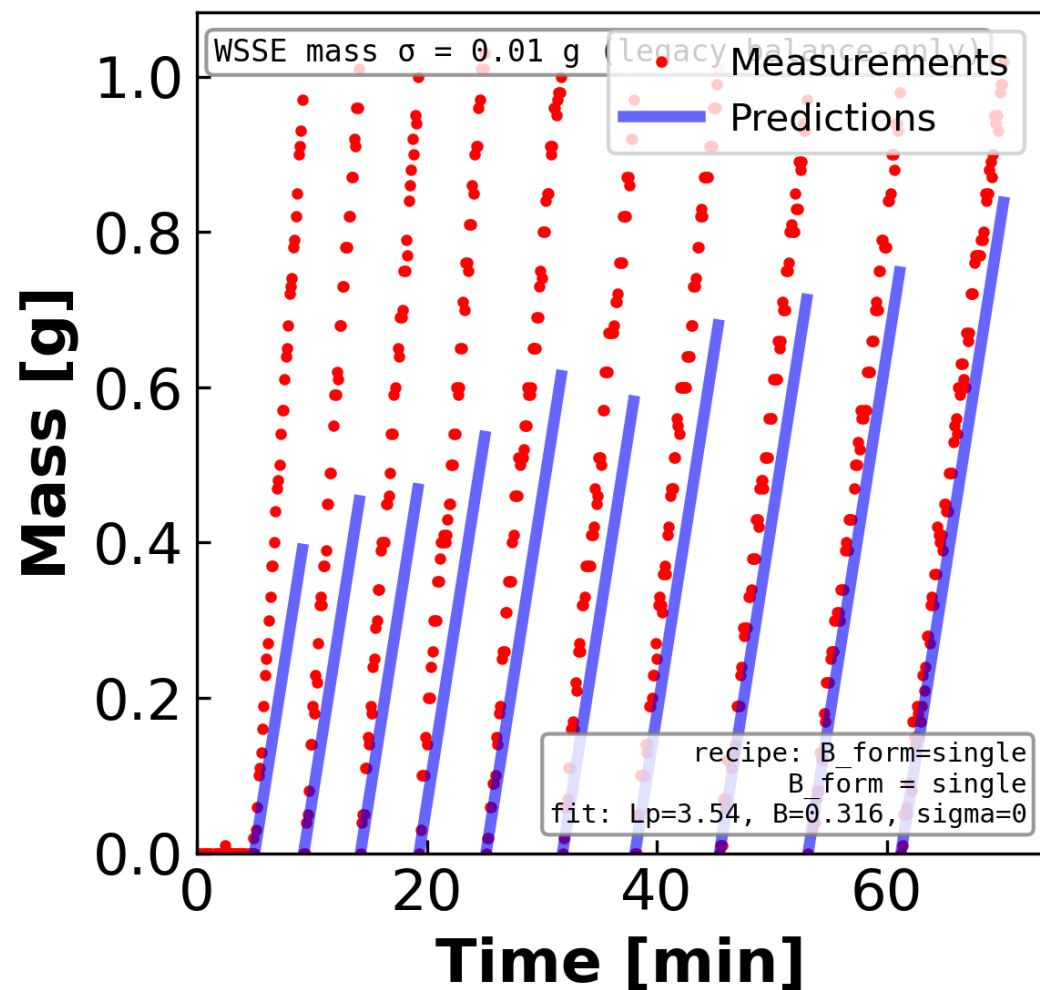
Backup — Figure 2. Measured and model-predicted time profiles for a representative...



Backup — Figure 3. Measured and model-predicted time profiles for a representative...

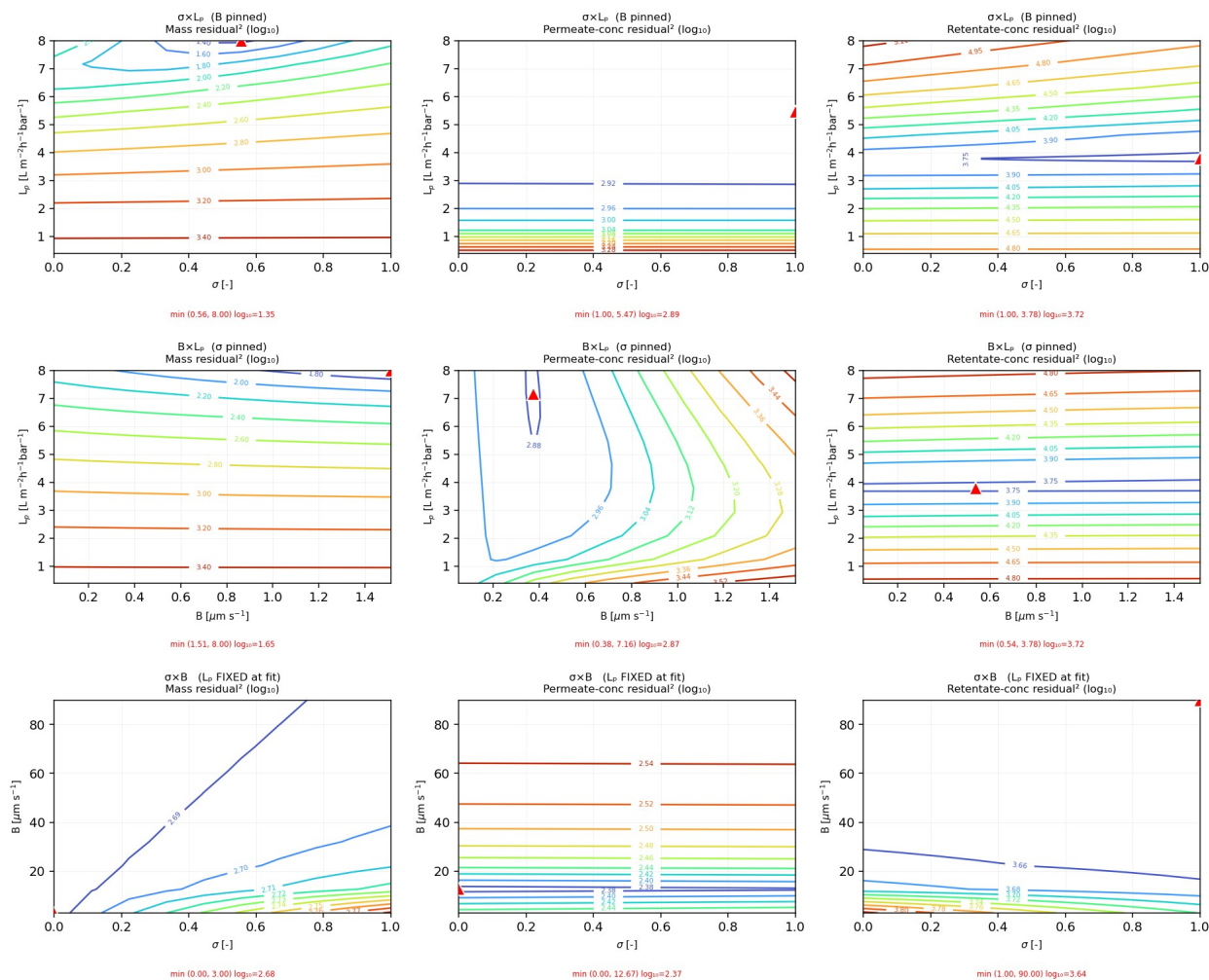


Backup — Figure 4. Measured and model-predicted time profiles for a representative...



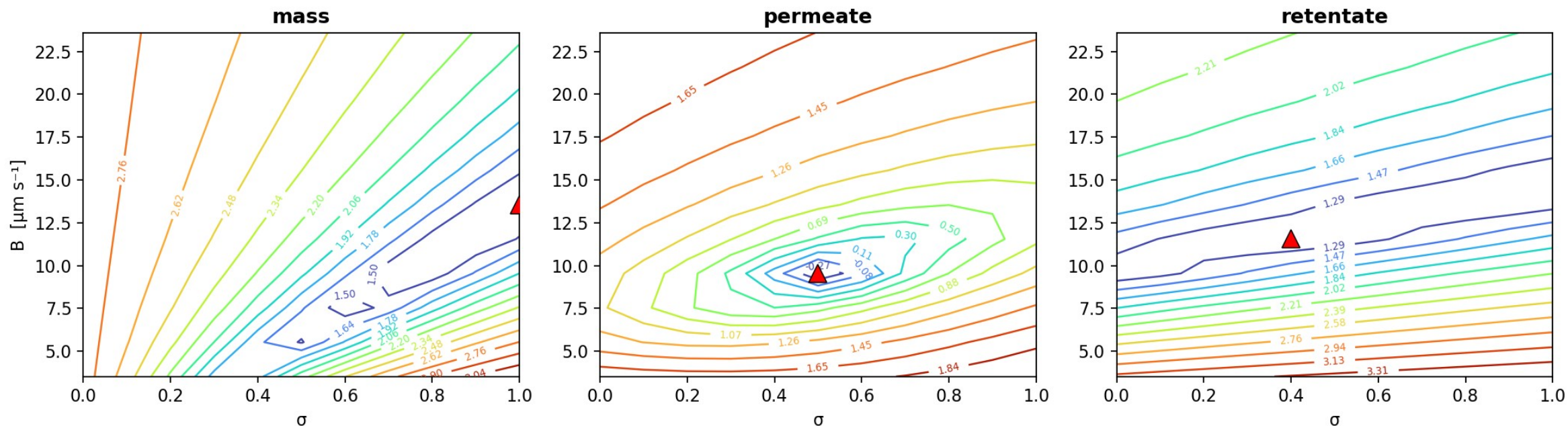
Backup — Figure 5. Log-transformed residual-squared contours for a representative...

CaCl₂ (MC2.05.07.24_CaCl2) · DATA1-style identifiability slices (DATA3 experiment): L_p pinned by mass (rows 1-2) $\Rightarrow$ focus on σ -B at fixed L_p (row 3)



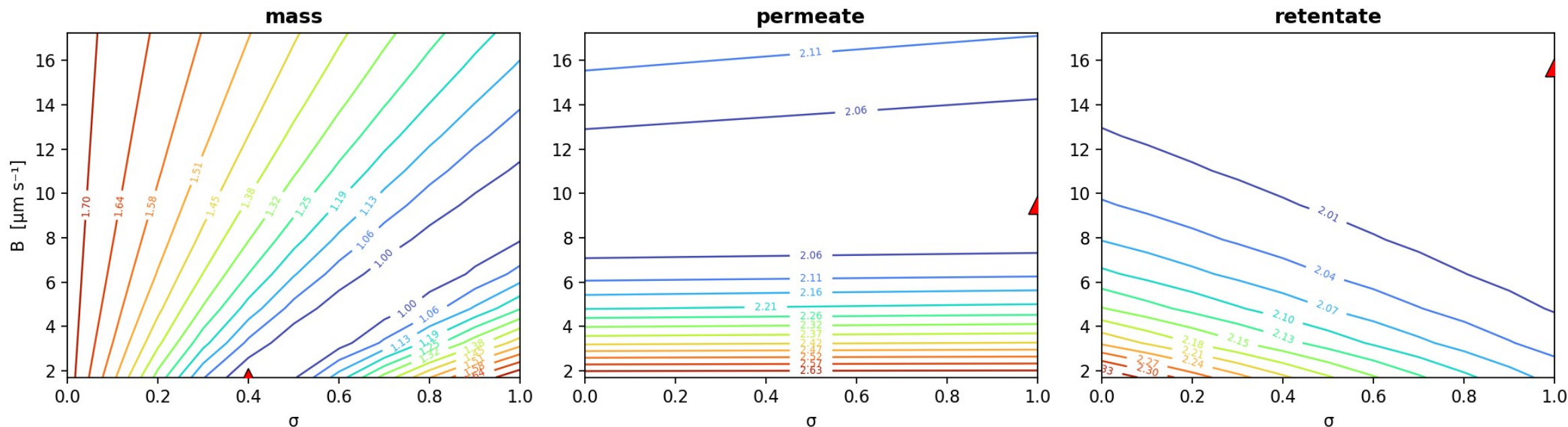
Backup — Figure 6. Identifiability of the reflection coefficient σ and the solute...

diluting NaCl (MC3.07.22.24) — $\sigma \times B$ identifiability, hydraulic permeability fixed at its identified optimum $L_p = 8.98 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$



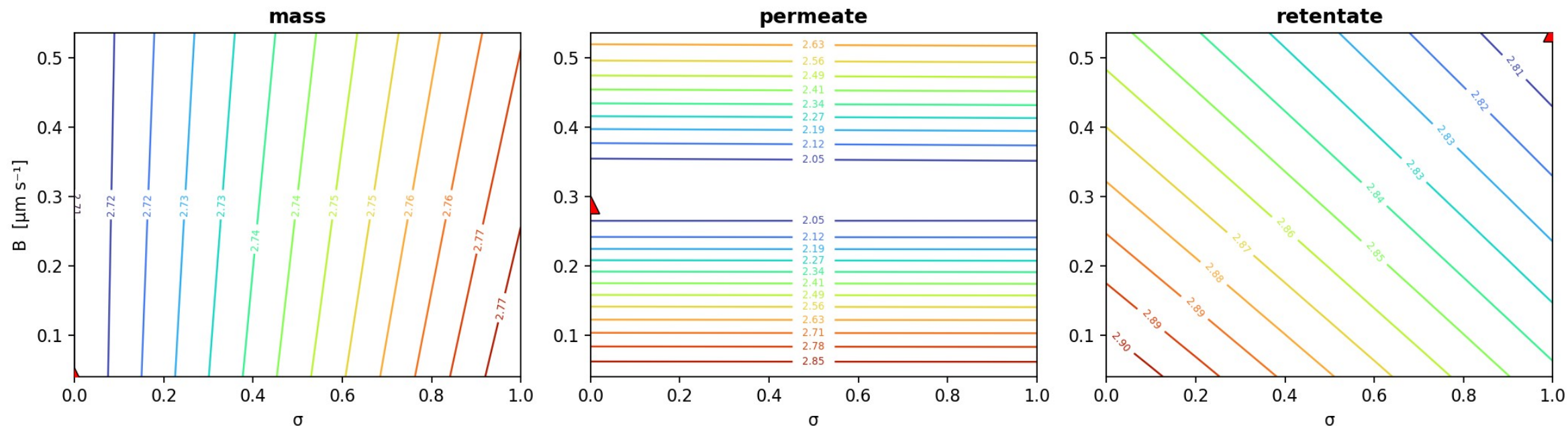
Backup — Figure 7. Identifiability of the reflection coefficient σ and the solute...

concentrating NaCl (MC2.05.07.24) — $\sigma \times B$ identifiability, hydraulic permeability fixed at its identified optimum $L_p = 9.91 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$



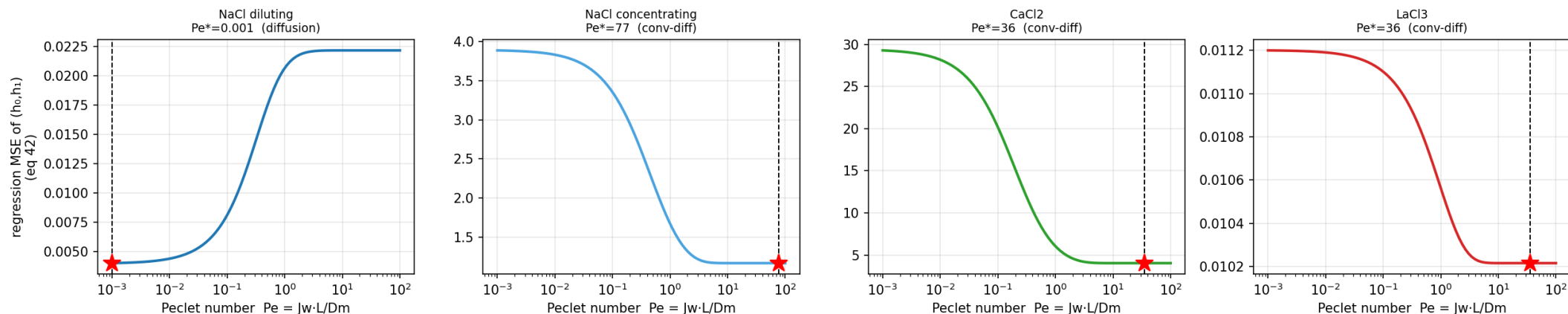
Backup — Figure 9. Identifiability of the reflection coefficient σ and the solute...

concentrating LaCl_3 (MC2.05.21.24) — $\sigma \times B$ identifiability, hydraulic permeability fixed at its identified optimum $L_p = 3.54 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$

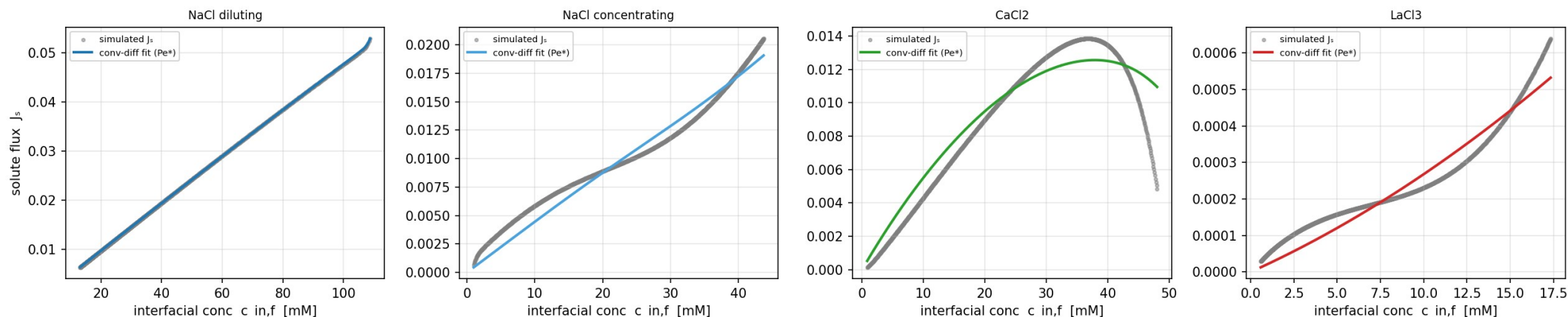


Backup — Figure 10. (Left) Mean squared error of the convection-diffusion...

DATA3 Peclet analysis ($\triangleq$ DATA2 Fig 8A): regression MSE vs Pe — a minimum at $Pe \geq 1$ means convection+diffusion, i.e. B is concentration-dependent (not the $Pe \rightarrow 0$ constant-B limit)

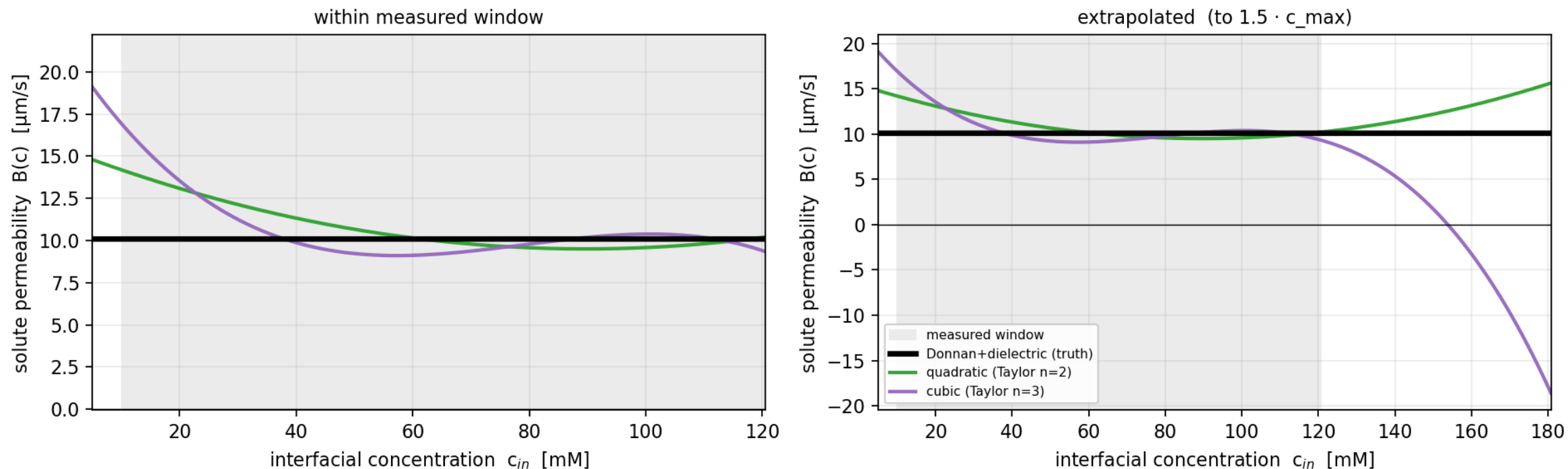


DATA3 ($\triangleq$ DATA2 Fig 7C): simulated J_s vs the exact convection-diffusion solution at Pe^*



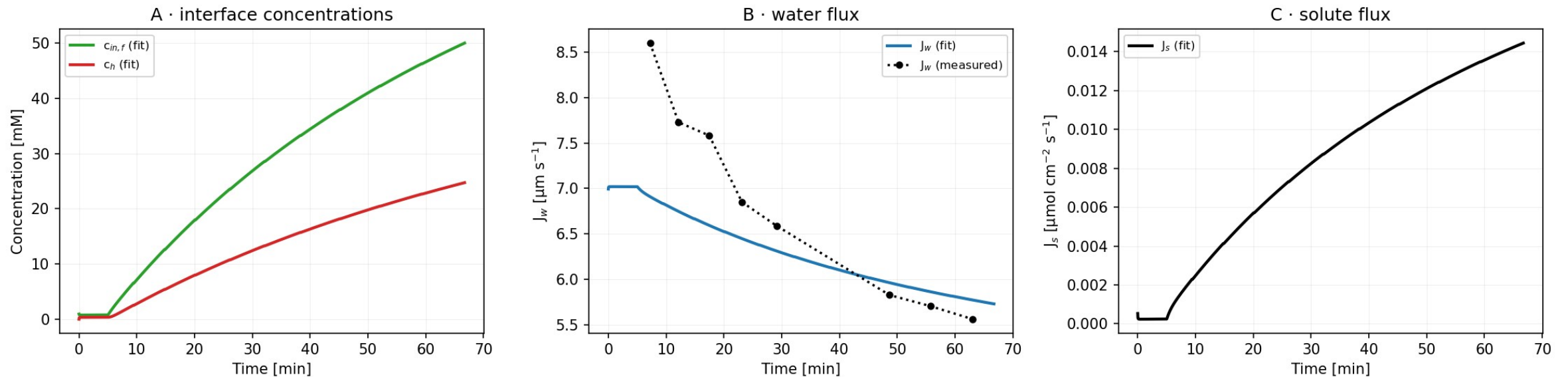
Backup — Figure 15. Solute permeability B as a function of interfacial concentration for...

NaCl (MC5.07.23.24) — Taylor-series $B(c)$ vs mechanistic Donnan, REAL fitted parameters
Donnan is the bounded 'truth'; polynomials are Taylor approximations that diverge on extrapolation



Backup — Figure 16. Forward-simulated membrane-interface concentration $c_{in,f}$ and bulk...

CaCl₂ — concentrating · MC2.05.07.24_CaCl₂ · concentration & flux time-series



Backup — Shifting Transport Dominance (Extended Nernst–Planck)

$$J_i = v_s \cdot C_i \cdot K_{i,v} - D_i \cdot \nabla C_i - z_i \cdot C_i \cdot D_i \cdot (F/RT) \cdot \nabla \psi$$

Convection

Diffusion

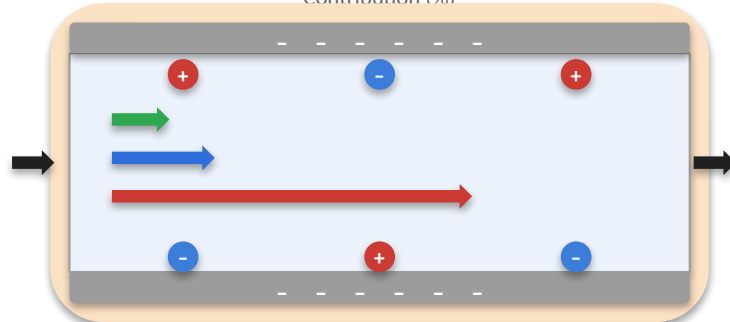
Electromigration

Low Molarity (~0.01–0.1 M)

Contributions to flux (J_i)



Contribution (%)



Pore cross-section — fixed (-) wall charges, mobile ions (+/-), flux arrows sized by contribution; flow is Feed → Permeate (left → right)

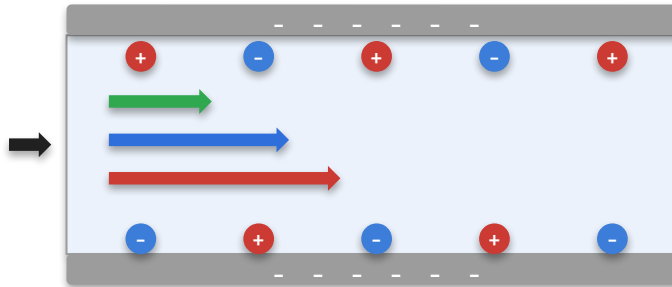
- Electromigration (dominant)
- Diffusion (secondary)
- Convection (minor)

Moderate Molarity (~0.1–0.5 M)

Contributions to flux (J_i)



Contribution (%)



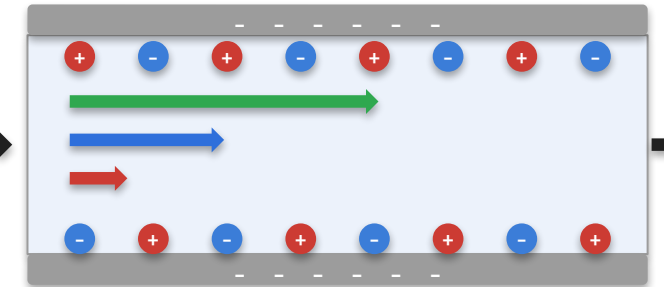
- Convection (increasing)
- Diffusion (comparable)
- Electromigration (significant)

High Molarity (~0.5–2+ M)

Contributions to flux (J_i)



Contribution (%)



- Convection (dominant)
- Diffusion (secondary)
- Electromigration (minor)



Role of fixed negative charges & the Donnan potential. Negatively charged pore walls repel co-ions and attract counter-ions, creating a Donnan potential ψ_D that drives electromigration. At low molarity the low ionic strength magnifies ψ_D , making electromigration the primary driver; as molarity rises, charge screening by abundant ions shrinks ψ_D , so convection increasingly governs transport.